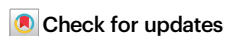


Many small climate change impacts presage rapid population extinction in a common iconic bird

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Population declines in common bird species abound, but it is unclear if these declines presage extinction. The consequences of climate change are particularly elusive, as there are typically both positive and negative effects on the seasonal- and life-cycles. We examine climate sensitivities of a population of a common and loved Australian bird, the superb fairy-wren *Malurus cyaneus*, with three decades of year-round fine-scale measurement of reproduction, survival and immigration. Population growth has been affected by many aspects of climate, but declines are most strongly linked to low fecundity in dry springs and reduced survival during unusually warm winters and preceding hot summers. Our comprehensive integrated population model is effective at summarizing these effects and predicting recent population volatility, suggesting it will be useful for assessing future population fluctuations and viability. A high risk of population extinction in the next 30–40 years is predicted. Accumulation of many small effects of climate change causes this decline. Hence a common species of Least Concern is in imminent danger, with virtually no time or way to prevent it. Such predicaments can only be detected in very well-studied species, and superb fairy-wrens may be the canary in the coal mine.

Extinction risk is highest for rare species, as their small number and restricted distribution make them vulnerable to local environmental change and chance events. However, this does not mean that common species are safe. Historically, some common species have gone extinct very rapidly. Notably, the passenger pigeon (*Ectopistes migratorius*) was once the most abundant bird species (~3–5 billion individuals) but went extinct only a few decades after the first concerns about their declining numbers were raised¹. Concerningly, ample evidence has emerged that many common species are rapidly declining^{2–4}. Whether

and how fast the current decline of common species will lead to extinction is unclear. However, this question is urgent, as the extinctions of common species are likely to have disproportionate impacts on ecosystems^{5,6}. Furthermore, rapid declines leave little time for conservation action to prevent extinction and its cascading impacts.

Most historic rapid extinctions have been caused by human-driven overexploitation, deliberate or inadvertent introduction of predators, and habitat loss². The increased tempo of anthropogenic climate change is exacerbating some of these threats, but also posing

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new threats^{7,8}. However, whether climate change can also lead to rapid extinction of common species is unknown and more challenging to study. This is because climate (change) affects many physiological and behavioural traits, as well the environment (resources, predators, phenology, habitat). Consequently, climate change can affect a range of vital rates across the seasonal- and life-cycles of organisms in positive and negative ways, making it hard to identify its overall impact. This problem is exacerbated by the fact that most studies only monitor species during part of the year, meaning effects could be overlooked or obscured by opposing effects in non-monitored parts of the seasonal cycle. Finally, the timescale of much research is insufficient to identify the impacts of extreme events, a key component of climate change⁹. In this study, we show that many small climate change effects can accumulate to drive rapid population extinction in a common iconic bird.

Results and discussion

Superb fairy-wrens

We used field data on a common resident songbird of south-eastern Australia, the superb fairy-wren (*Malurus cyaneus*), to first identify the demographic pathways whereby climate affects population dynamics via vital rates, and then use this to quantify future extinction risk under climate change. Fairy-wrens are an ideal study species because (i) their vital rates can be measured precisely with colour banding and intense survey^{10,11}; (ii) previous studies found clear climate impacts on their life history traits (e.g., breeding phenology¹¹ and chick body mass¹²) and vital rates (e.g., adult survival rate in the non-breeding season¹⁰); and (iii) declines have been observed in some populations of this species^{13,14}. We have studied reproduction and survival of a fairy-wren

population in the Australian National Botanic Gardens, Canberra, for three decades, achieving year-round fine-scale censuses that characterized the weekly timing of mortality, all nesting attempts, and all immigration¹⁵. The study area is an elevated inland site with strong seasonal variation in temperature, and alternation between drought and more mesic conditions. The population fluctuated throughout the 1993–2022 study period, though it generally declined until a burst of growth in the final two years (Fig. 1A).

Male numbers were consistently greater than female numbers (Fig. 1A). This male excess is due to cooperative breeding, as males are tolerated on their natal territory, and may help the breeding pair through provisioning of offspring and nest defence for up to 6 years¹⁶. If males become dominant, they almost always do so on their natal territory or on an immediately adjacent territory. By contrast, juvenile females must disperse in their first year to obtain a breeding vacancy, and often move over large distances^{17,18}. Females achieve this dispersal in two ways. Females fledged before mid-December typically disperse before the end of the breeding season to a subordinate role in a foreign group. Females fledged after that date overwinter in their natal territory. All young females enter a scramble for a breeding vacancy at the start of spring, except for early dispersers that have already inherited the territory on the death of the incumbent. Early dispersers are also advantaged over overwintering natal birds because they can form an alliance with a helper male and fission the territory in which they have settled, and are much more likely to obtain a vacancy near where they have overwintered¹⁷. Females that do not obtain a vacancy in the breeding season after they fledge usually die, generating the male bias in adult sex ratio.

These social influences led us to consider three phases of the annual cycle (Fig. 2) during which climate impacts may occur. I. the recruitment phase (mid-November to March), where territories accumulate extra members through reproduction and immigration of young females; II. the non-breeding phase (Apr–Jul), where there is little movement but decreased survival; and III. the scramble phase (August to mid-November), where the number of breeding territories is settled through fission and fusion of existing territories, and the replacement of breeding females that have died.

Integrated population model

Integrated population models (IPM) combine individual- and population-level data into a single statistical model that also describes their interrelationship governed by demography and population dynamics¹⁹. We first developed an IPM to estimate all vital rates and annual population growth rates (PGRs). To facilitate understanding, we only present a general non-technical description of the IPM here; a technical description is in “Methods”. Our model considered five different stage classes to account for substantial differences in (climate impact on) vital rate among types of individuals (Fig. 2). We considered two sexes (male and female) each with two age classes (juvenile and adult, the latter being members of groups attending a nest). Juvenile females were split further into two stage classes that differed in their recruitment prospects: natal females that stayed in their natal territory at the end of the recruitment phase, and non-natal juveniles that moved out of natal territory during the recruitment phase.

Transitions between these stage classes was governed by six stage-specific vital rates: apparent survival rate ϕ , the probability that an individual survived (and remained in the study population) over a given phase; immigration rate δ , the number of immigrants in a given phase relative to the number of adult females (or males) at the end of the scramble phase; fecundity γ , the number of juveniles produced by each female that survived and stayed in the study area in the recruitment phase; offspring sex ratio τ , the proportion of females among juveniles; female natal philopatry rate π , the proportion of juvenile females produced in the study population that stayed in their natal territories in the recruitment phase; and the probability of males being

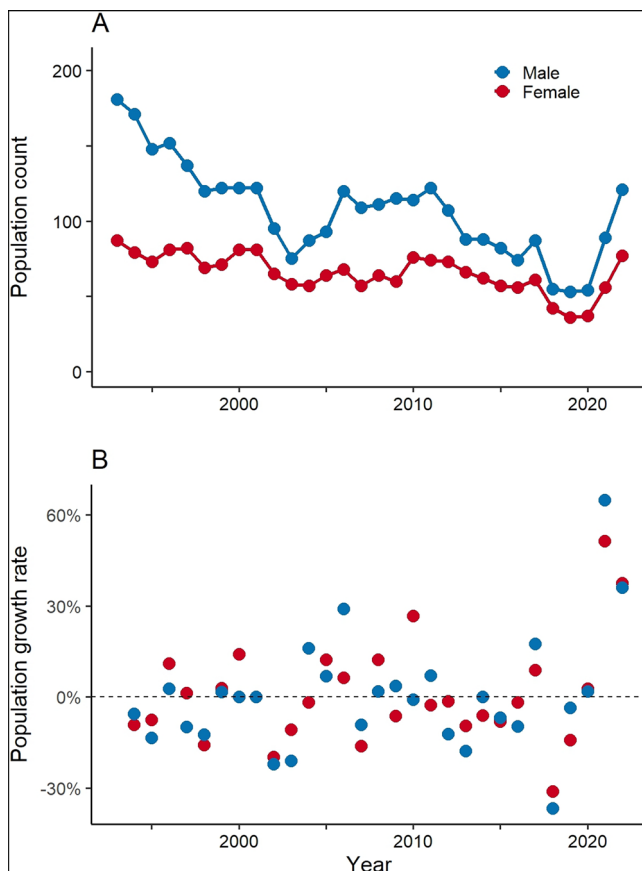


Fig. 1 | Population dynamics of male and female superb fairy-wrens. observed number of individuals in population counts at the end of the scramble phase (15 November) (A), and annual population growth rate (B).

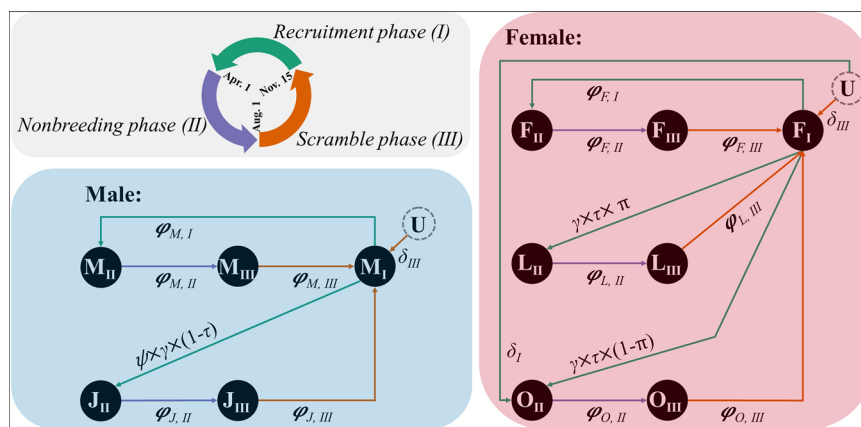


Fig. 2 | The life-cycle diagram used to model the dynamics of superb fairy-wrens. The top left panel shows three phases of the annual cycle. The red and blue panel show the five stage classes of female and male individuals in the population: F: adult females; L: natal juvenile females (who remain in the territory where they were born for their first winter); O: non-natal juvenile females (who have

immigrated to a subordinate role before their first winter); M: adult males; J: juvenile males (which are virtually all natal). 'U' denotes unobservable individuals outside of the study area who are the source of immigrants. Arrows indicated the transition probabilities among stage classes between phases of the annual life cycle (i.e., vital rates; see details in the main text).

the dominant in their territory ψ (detailed algebraic definitions in “Methods”). Furthermore, vital rates (and climate impacts on them) were specific to each annual phase. Finally, each vital rate was modelled as a binomial or Poisson random variable to account for uncertainty due to small population size.

Demographic pathway of climate impacts on population dynamics

To identify the main demographic pathways via which climate affects population dynamics, we conducted a retrospective path analysis^{20,21} to quantify how climate variables affect annual PGR through vital rates. This effect of climate on annual PGR via each vital rate is given by the product of how much that vital rate is affected by the climate variable and how much the change of annual PGR is affected by a change in vital rate (Supplementary Fig. 4). Based on detailed previous studies on this population, we explored the effects of five climate variables on all relevant vital rates in the different annual phases (Supplementary Fig. 1): spring rainfall (Aug–Nov)^{22,23}, summer maximum temperature (Dec–Feb)^{10,11}, summer rainfall (Dec–Feb)¹¹, and winter minimum and maximum temperatures (Apr–Jul)¹⁰. Vital rates were also density-dependent, but as this did not significantly affect our conclusions; we have confined the description of density-dependence to “Methods”.

We found 11 demographic pathways via which climate affects population dynamics of superb fairy-wrens (Fig. 3). More spring rainfall in a given year was associated with a higher annual PGR through increased fecundity and immigration rate of juvenile females in the recruitment phase (Fig. 3 and Supplementary Figs. 2A, 2B and 4). The breeding season is very long, females can produce up to four successful broods, and the clutch size can also vary. As reported previously, prolonged breeding, increased clutch sizes and promoted fecundity are strongly positively associated with high spring rainfall^{11,22,23}. Hence both the number of natal and early-dispersing young increased after high rainfall. In extreme droughts, recruitment can be severely limited and the population therefore declines as breeding females die and are not replaced. This often leads to fusion of neighbouring territories¹⁷.

The effects of temperature on annual PGR are also important. Population growth was limited by above-average winter and summer maximum temperatures, and by below-average winter minimum temperatures, mostly through reduced adult and juvenile survival rates in the non-breeding phase (Fig. 3 and Supplementary Figs. 2–4). Increased maximum temperatures in the winter worsen

the severity of an annual episode of reduced survival that occurs just after the winter solstice, when temperatures are coldest¹⁰. The survival may be particularly low when warm conditions are followed by cold snaps¹⁰. Finally, there appear to be deleterious effects of warm summers on survival in the subsequent year¹⁰. All of the effects on adults reported in a previous study¹⁰ are recovered in the current study with more years of data (Fig. 3). Juveniles similarly may be affected by temperatures as adults, especially considering that juvenile survival is usually affected by weather more than adult survival²⁴.

Future population extinction risk under climate change

We assessed the fate of the population under continued climate change. To first validate the predictive value of our model, we fitted a model to our data from 1993 to 2015 and determined how well it predicted the population changes from 2016 to 2022. These last years of the study are particularly suitable for validation as they included climate extremes, and unprecedented volatility, including both the sharpest population decline and the two years when population growth was greatest (Fig. 1B). This cut-off was also convenient because 2016 was the hottest year on record in south-eastern Australia, making it a critical test case of our accuracy in projecting to novel extreme conditions. Both the projected sharp decline and subsequent recovery were qualitatively in agreement with the observed population fluctuations from 2016 to 2022 (Fig. 4A). Although the model slightly underestimates the depths of the recent decline and the subsequent rebound (see next section for more discussion), the characterization of both these events supports the view that our IPM could project the population size reliably.

Next, we used the model that was validated and parameterized with historical field data to projected annual population size from 2022 to 2100 by using forecasted climate variables under scenarios of low, intermediate and very high greenhouse gas emission (respectively, IPCC Shared Socioeconomic Pathways 1–2.6, 2–4.5 and 5–8.5²⁵). The forecasted maximum and minimum temperatures in summer or winter increased in most emission scenarios and the rate of warming amplified from low, intermediate to very high emission scenarios (Supplementary Table S1). Although spring rainfall generally showed no directional change across years in any of the emission scenarios (Supplementary Table S1), its effect on recruitment can still compound the deleterious effects of temperature on survival. In addition, as a control, we also projected population dynamics under a scenario with no further increased emission-driven

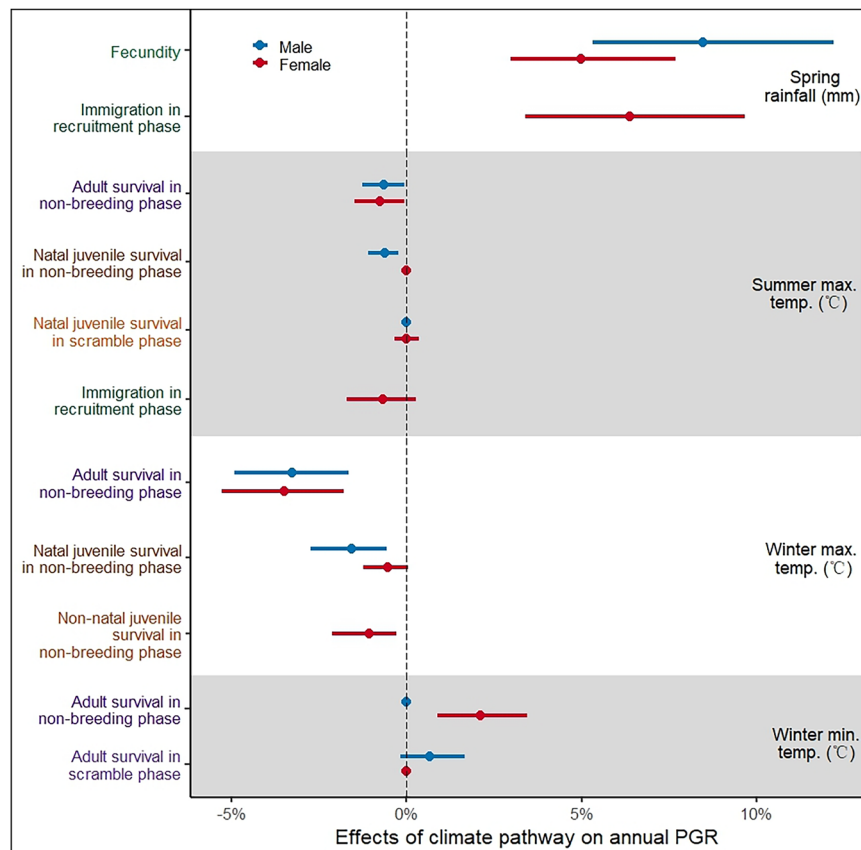


Fig. 3 | Summarized results of the retrospective path analysis of climate on annual population growth rate (PGR) mediated through the effect of vital rates in female and male superb fairy-wrens. Effects (mean $\pm$ 95% credible interval; estimated using 6000 posterior samples generated by the integrated population model) of daily average spring rainfall, summer maximum temperatures, winter maximum temperatures, and winter minimum temperatures on annual PGR. As an

example, spring rainfall affected both fecundity and female immigration rate in the recruitment phase, and consequently contributed to the change of annual PGR. If daily average rainfall in spring increases by 1 mm, female population size increases ~5% through increasing fecundity. Dots without credible intervals on the dashed lines indicate that the effect of a given climate variable on a given vital rate was excluded in the variable selection procedure.

directional climate change (see “Methods”). To appreciate the impact of climate change on population dynamics relative to other sources of variability and uncertainty, our ensemble projections averaged over all 21 climate models and estimation uncertainty in all model parameters, and included demographic and environmental stochasticity (see “Methods”).

Under the scenario of no further climate change, superb fairy-wren population size was forecasted to fluctuate with a low risk of population extinction (22.9% probability of female population size decline to zero in 2100; Fig. 4B). In contrast, although the projected population trajectories differed somewhat among different greenhouse gas emission scenarios, climate change in general led to dramatic and rapid population decline and consequently all projections went extinct, even in the low emission scenario (Fig. 4C, D). Additionally, the 50% risk of population extinction occurred progressively earlier from low, intermediate to very high emission scenarios. Under the intermediate and very high emission scenarios, the fairy-wren population size was forecasted to go extinct between 2059 and 2062. This implies that we have only 3–4 decades to prevent likely local extinction, as current global emission reductions exceed the low scenario²⁶.

Are there avenues for population rescue?

The alarming projections of extinction raise the question of whether any actions or factors are likely to prevent or delay extinction. Three possibilities warrant discussion.

The first is genetic adaptation under selection. Although our population projection assumes that the effects of climate on vital rates in the future will be the same as in our study period, evolutionary rescue in vital rates may occur and reduce the risk of extinction. For example, after an extreme cold event, green anole lizards (*Anolis carolinensis*) become more resistant to cold and show gene changes that are relevant to thermoregulation²⁷, which could help them to survive future cold winters better. A recent synthesis of long-term data from wild birds and mammals also suggests that the potential for rapid adaptation was greater than hitherto believed, but the genetic variance for fitness in superb fairy-wrens was at the lower end of the species considered²⁸. It has also recently been argued that changes in the direction of selection associated with climate fluctuations can impede evolutionary rescue²⁹.

The second is the immigration of warm-adapted genotypes. The breeding distribution of superb fairy-wrens spans more than 20 degrees of both latitude and longitude, and most of Australia's altitudinal range^{30,31}. The population under study experiences a continental climate exacerbated by relatively high altitude, meaning it experiences colder winters and drier conditions than in much of the species range. However, there is likely connectivity to populations adapted to warmer (and possibly drier) environments. The extreme philopatry of male fairy-wrens retards gene flow, but the obligate dispersal of young females has been shown to introduce novel genotypes into our study population³², so rescue by gene flow is possible.

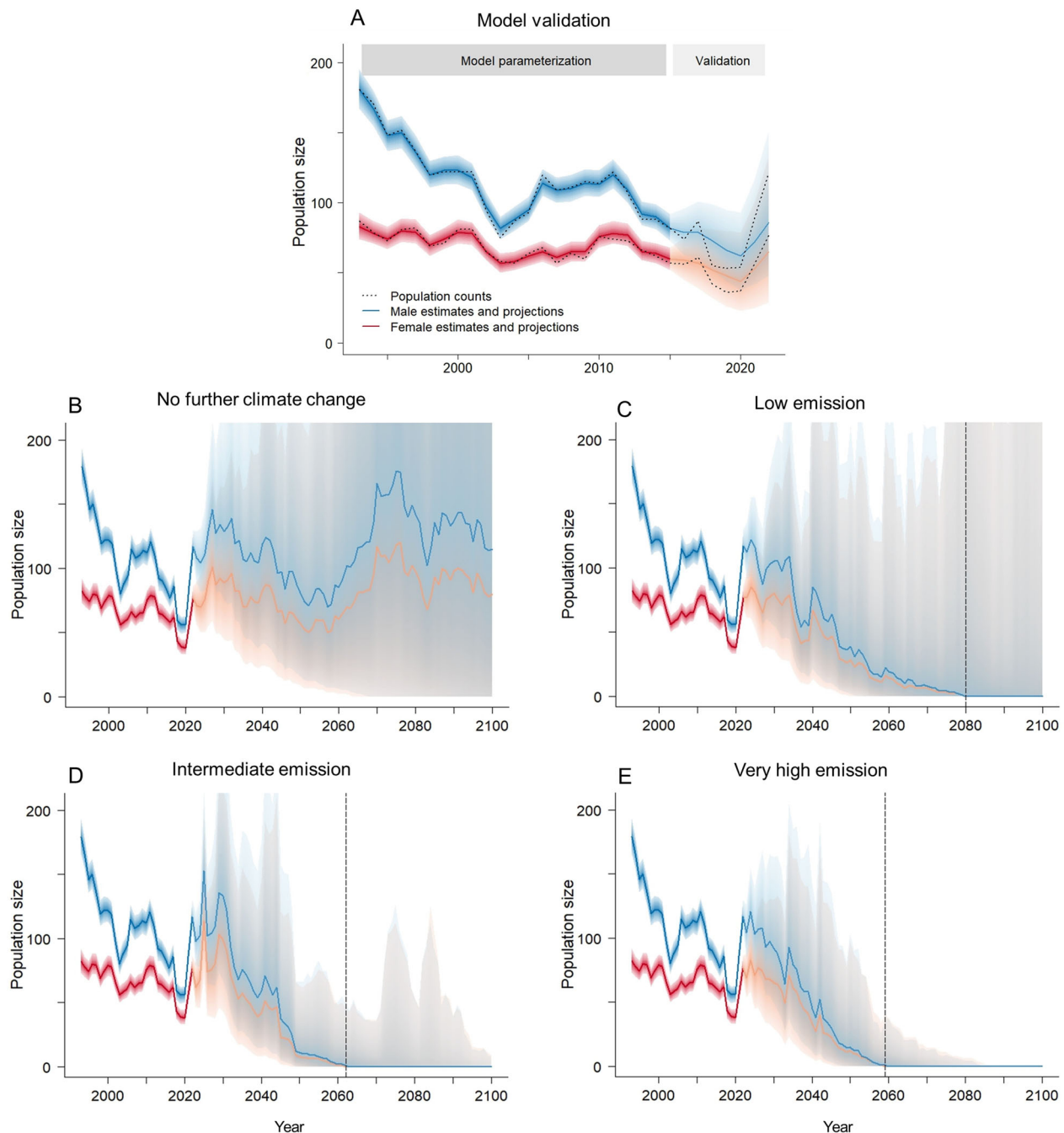


Fig. 4 | The estimated historical and projected population sizes of female and male superb fairy-wrens. in each year at the end of the scramble phase (15 November). Validation of the ability of an integrated population model (IPM) to predict population size (A), and the projected population size under the scenario of no further climate change (B; i.e., climate will not further change due to emissions) and three climate change scenarios: *low* (C), *intermediate* (D) and *very high* (E) greenhouse gas emission scenarios. The latter three sets of forecasted climate data were obtained from 21 underlying global climate change models of the Coupled Model Intercomparison Project Phase 6, the influences of which we summarized using ensemble projections. In panel (A), observed numbers are shown by dashed lines (see also Fig. 1A). The bold colours represent the estimated median number of

individuals with the corresponding shaded area depicting the 80% uncertainty strip by the IPM parameterised with the climate and demographic data from 1993 to 2015 in panel (A) (1993 to 2022 in panels B–E). The right two thick lines are the projected median population size from 2016 to 2022 in panel (A) (2023 to 2100 in panels B–E) with the corresponding 80% uncertainty gradient (population prediction interval), based on the IPM parameters from 1993 to 2015 in panel (A) (1993–2022 in panels B–E). Hence, the IPM is fully parameterised until 2015 in panel (A) (until 2022 in panels B–E) but predictive thereafter. In panels (C–E), the dashed vertical line in each panel is the year in which the probability of extinction is 50% (i.e., the median population size of the female equals zero).

Third, climate change affects all the species in a habitat. We have evidence that during the period over which superb fairy-wrens have declined there has been a far steeper decline in numbers of their most significant nest predator, the pied currawong (*Strepera graculina*). Currawong territories are large and hence the population is more

difficult to enumerate, but has potentially declined from a peak of more than 20 territories to just 5 in 2020. Nest predation, the most significant cause of nesting failure, has declined accordingly, which may help explain the slight underestimation by our IPM of the population surge in the last two years reported here. Superb fairy-wrens

may thus be rescued by climate-driven reductions in predators, but this hypothesis awaits further testing.

Accumulation of many small climate change impacts

Despite some potential for natural rescue, the dire model projections cause considerable alarm. Population extinction is likely to happen very fast; within the next 30–40 years in the intermediate and very high emission scenario. This leaves little time for conservation actions, a realization that is striking given that superb fairy-wrens are very common in a range of habitats and have always had a conservation status of Least Concern³³. We previously reported that some aspects of their life-cycle are influenced by climate change^{10,11}, but this has not led to concerns about their future viability thus far (also not by us, the researchers). Only by integrating the many impacts of climate change across the seasons and the full life-cycle, we found that although many individual impacts are small or moderate, together they are expected to have a catastrophic cumulative effect, even in the low-emission scenario.

We identified 11 demographic pathways via which a suite of climate variables operate at different times of the year, with different time-lags, and differential impacts on males and females. This knowledge is not only important for making reliable projections but also for identifying the nature and limits of conservation actions, as many demographic pathways make it hard to come up with solutions. Mitigating climate change impacts is challenging, as it cannot be solved locally and can only be buffered or compensated by improving other environmental conditions. However, mitigating a wide diversity of impacts is even more challenging. Given that most conservation success stories involve species in which a single demographic cause of decline was resolved^{34,35}, we should be modest and realistic (i.e., pessimistic) about what conservation can achieve in complex cases like superb fairy-wrens, especially within such a short time frame.

Population decline and extinction of common species

The deleterious effects of climate change, and particularly of increased winter temperature on our population of superb fairy-wrens is surprising, as a priori an increase in winter temperatures should move these endothermic birds closer to their thermoneutral zone and lessen their energetic burden. Although we lack the relevant data, which would be extremely expensive to obtain, we suspect the problem posed by sudden warm spells followed by cold spells¹⁰ may negatively affect the ectothermic arthropod prey exploited by the birds, possibly by provoking emergence and activity at the wrong time of year. In that sense, our data resemble the large reduction in populations of common bird species in both Europe and North America^{3,4}, where the greatest declines are in species that are wholly insectivorous or feed their young on arthropods^{36–38}. Like bird populations, some arthropod declines have been caused by habitat simplifications through agricultural intensification, and the use of novel insecticides^{39–41}. However, there is also evidence that huge arthropod declines can occur in environments that are not subject to either of these influences. These have been attributed directly to global warming⁴². The decline in fairy-wrens we report on here is not in agricultural habitat, but their habitat is experiencing the deleterious effects of warming.

Our studied population of superb fairy-wrens is predicted to crash under all climate change scenarios. This result echoes the recent reports that climate change will limit population growth and lead to local extinction of wild animals^{43,44}. For example, a recent study on African wild dogs (*Lycaon pictus*) found that climate warming will lead to population collapse in both intermediate and very high emission scenarios⁴⁵. Although it is well-established that climate change threatens the population persistence of wild animals^{8,46}, our study uniquely provides demographic evidence that climate change poses an urgent threat to common species. As ecosystems are likely to be disproportionately impacted by the extinctions of common species^{5,6},

more conservation attention to common species is needed in the future viability studies of wild animals.

Our study highlights the value of year-round monitoring and comprehensive analysis of all vital rates throughout the seasonal- and life-cycles. We expect it to be a common feature in many species that climate change affects them in many ways. However, most species have not been comprehensively studied year-round for decades, meaning that such effects also act during parts of the year when researchers are not in the field, and thus will often not be identified. In that sense, superb fairy-wrens may be the canary in the coal mine.

Methods

Ethics

The study period was covered by multiple permits, including the approvals from the Animal Experimentation Ethics Committee of the Australian National University (to A. Cockburn; A2022/20 (approved on 10 August 2022), A2019/23 (approved on 9 August 2019), A2016/22 (approved on 10 August 2016), A2013/27 (approved on 9 August 2013), F.BTZ.42.10 (approved on 10 August 2010), F.BTZ.06.07 (approved on 10 August 2007), F.BTZ.23.04 (approved on 10 August 2004) and F.BTZ.16.01 (approved on 10 August 2001)) and the Australian National Botanic Gardens (to A. Cockburn and H. L. Osmond; 2020/21-1 (approved on 27 June 2020), 2013/4-1 (approved on 9 December 2013), 2008/9 (approved on 1 January 2008), 2006/1651 (approved on 1 January 2006) and 2004/1549 (approved on 1 January 2004)). In addition, A. Cockburn held permits from the Animal Experimentation Ethics Committee of the Australian National University between 1993 and 2001, and both A. Cockburn and H. L. Osmond held approvals from the Australian National Botanic Gardens from 1993 to 2004.

Overall approach

Our two aims are (i) to quantify how climate affects population dynamics via vital rates in a population of superb fairy-wrens (*Malurus cyaneus*), and (ii) to project population viability under future climate change. To address the first aim, we developed an integrated population model (IPM) to estimate vital rates and annual population growth rate (PGR) throughout the annual cycle. Subsequently, we conducted a retrospective path analysis to measure the effects of climate on annual PGR through each vital rate. To address the second aim, we extended the previous IPM by including the historical effects of climate variables on each vital rate and combined it with forecasted future climate downscaled to the study area, to project population viability under different future climate change scenarios.

Model system

Superb fairy-wrens are a common resident passerine of south-east Australia. A population of fairy-wrens in the Australian National Botanic Gardens, Canberra (35°16'S, 149°6'E), has been studied continuously since 1988. The study area was expanded substantially in 1991/1992, therefore we focus here on the data from 1993 to 2022. The study area encompasses a total of ~60 hectares divided almost equally between plantation of Australian native shrubs and trees, and natural *Eucalyptus* woodland. The study population is a sample of a much larger population with free movement across its boundaries to unmonitored territories; we included a given territory if >50% of the territory was within the study area. There were between 36 – 87 fairy-wren territories in a given year (Fig. 1A).

Superb fairy-wrens form socially monogamous pairs and live in year-round territories. All birds in the population are individually colour-banded, either as nestlings if they were born in the study area (at 5–7 days old), or as juvenile or adult immigrants caught by mist-netting. We aimed to observe every bird at least weekly throughout the year. If a bird was not seen on routine censuses, we made further attempts to find it, resulting in near-complete detection rates¹⁰. The breeding pair on each territory can either breed unassisted or be helped by up to 5

male helpers in this cooperatively breeding species¹⁵. During the breeding season (Aug–Mar¹¹) we monitored every nesting effort by every female at least every three days. Although the adult sex ratio is male biased, there is negligible bias in the sex ratio of hatchlings⁴⁷, and adult males and females have similar survival rates⁴⁸.

Male offspring are extremely philopatric birds occupying year-round territories, and generally live and die on their natal or an immediate neighbouring territory¹⁶. Unbanded males are very rare and largely occur on peripheral territories. Both dominant and subordinate males are sexually active in their first breeding season¹⁵. In sharp contrast, female offspring always disperse away from their natal territory in the first year of their life, often moving more than two territories (few over 5 km)^{17,18}. Females start breeding in the first breeding season after they fledge and thereafter only rarely move more than one or two territories¹⁷. Therefore, virtually all female immigrants are in their first year of life.

Aside from the first year emigration and immigration of females before they gain a breeding vacancy, the loss of an individual from our study population is generally associated with mortality. The death of a breeder male or female is usually associated with rapid replacement by a helper or a disperser, respectively. Occasionally, birds on the periphery are lost from the census population because of short-distance breeding dispersal or fission of breeding territories, but are censused again when the territory to which they have moved fuses with a censused territory, or they return when their mate dies. This affects just 2% of the 2088 individuals considered (45/2088), of which most (40/45) are females.

Climate data

We used data on daily weather from an Australian Bureau of Meteorology automatic recording station at Canberra Airport (~8 km east of the study area and at similar altitude; <http://www.bom.gov.au/climate/data>). As the study site has a highly seasonal continental climate, and extreme weather is likely to affect the vital rates, we explored the effects of mean daily maximum temperature and mean daily rainfall in the austral summer (from 1 December to the following 28 or 29 February; hereafter ‘summer maximum temperature’ and ‘summer rainfall’), and mean daily minimum temperature in the austral winter (from 1 April to 31 July; hereafter ‘winter minimum temperature’). The effect of mean daily maximum temperature in winter (hereafter ‘winter maximum temperature’) was also included since we recently reported that higher winter maximum temperatures were associated with higher adult mortality in the non-breeding season¹⁰. Additionally, we tested the effect of mean daily rainfall in the austral spring (from 1 August to 30 November; hereafter ‘spring rainfall’) on fecundity because more spring rainfall increases the availability of food through benefiting plant growth and increasing abundance of arthropods for breeding superb fairy-wrens¹⁶. Although climate at shorter time intervals within a season (e.g., the coldest month of the year) might be expected to have strong effects on vital rates, preliminary analyses using average daily temperature and rainfall summarized at monthly intervals resulted in similar or weaker effects compared with when climate conditions were averaged across whole seasons.

To generate future values of these five climate variables (spring rainfall, summer maximum temperature, summer rainfall, and winter minimum and maximum temperature) from 2022 to 2100, we obtained monthly maximum and minimum temperatures, and rainfall, from all available global climate change models (<https://ds.nccs.nasa.gov/thredds/catalog/catalog.html>) of the Coupled Model Intercomparison Project Phase 6 (CMIP6)²⁵ with a spatial resolution of 0.25° latitude-longitude (~28 km), and chose the model grid point in which our study area is located. For the greenhouse gas emission scenarios of each climate model, we considered three widely used Shared Socio-economic Pathways (SSPs): (1) an optimistic *low* scenario with peak emissions in 2020 which was based on SSP 1–2.6; (2) an *intermediate*

scenario with peak emissions in 2040 which was based on SSP 2–4.5; and (3) a worst-case unrestrained *very high* scenario which was based on SSP 5–8.5. In our projections, we randomly sampled climate time series from one of 21 global climate change models (Supplementary Table S1) that provided monthly maximum and minimum temperatures, and rainfall for all three greenhouse gas emission scenarios. We did not consider intra-model uncertainty in our projections, as it was not available for all 21 models and is generally much smaller than the variability among climate models^{49,50}. The forecasted climate variables varied considerably among climate models for a given greenhouse gas emission scenario, but the tendencies for changes in the projection period are similar. As ensemble approaches tend to have better forecast performance⁵¹, we therefore focus on ensemble projections. To facilitate the understanding of the change of annual climate and seasonal climate variables in each climate change scenario, we summarized the change in each climate variable at both global and local (Canberra) scales in Supplementary Table S1.

For each set of the forecasted climate variables, their mean differed with the mean of corresponding historical climate variables in the overlapping years (2015–2021; $N = 7$), potentially due to the difference in scales: historical climate data from one local weather station versus forecasted climate data at a large grid cell level (0.25°×0.25° latitude-longitude; i.e., ~760 km²). To avoid the potential influence of different values of the mean between historical and forecasted climate data, and between different climate models for each greenhouse gas emission scenario in the forecasted climate data, we calibrated the values of forecasted climate variables from 2022 to 2100. Temperature variables and rainfall variable were each modified differently⁵². For each temperature variable, we subtracted the difference between forecasted temperature variable and historical temperature variable in the overlapping years. Whereas, for each rainfall variable, we multiplied the quotient of historical and forecasted rainfall values in the overlapping years. In this way, we retained the distribution and temporal change of the forecasted climate data.

Finally, to assess the impact of future anthropogenic climate change in above scenarios, we also generated a climate scenario of no further increased emission-driven directional climate change where the climate will not further change from 2022 to 2100. In the ‘no further climate change’ scenario, we generated values randomly for each forecasted climate variable from 2022 to 2100 using a Gaussian distribution with the mean and standard deviation of the corresponding historical climate variable from 1993 to 2021.

Integrated population model

To estimate vital rates and annual population growth rates (PGRs), we developed an integrated population model (IPM) which incorporated multiple types of data in a joint likelihood to estimate parameters. This allows for synthesizing data sources, efficient use and internal consistency of available information, and comprehensive error propagation throughout the population model¹⁹. Our IPM model included four submodels linked to the relevant datasets: (i) a population submodel utilising the population count data to inform historical changes in population size, (ii) a survival submodel using the individual capture-resighting data to estimate the apparent survival of different stage classes, (iii) a recruitment submodel utilising the recruitment data to estimate fecundity, and (iv) an immigration submodel using the immigration data to determine immigration rates. Below, we describe each sub-model and then the overall model.

Population submodel. The population submodel describes how female and male population size change throughout the annual cycle, and links the population count data with the population size variable in the model. Our model consists of a life-cycle structure that includes three phases in the year (*I–III*, Fig. 2) to accurately account for the fact that climate acts on different vital rates in different parts of the year.

Notwithstanding, for future projections we can focus on the dynamics at the end of the scramble phase (15 November) from one year (t) to the next ($t+1$), which means that our stage and phase-structured model can be summarized by a single equation for the dynamic of adult females or males from year to year Eqs. (1), (2). At this point, all juveniles have become adults or died, and our model simplifies to a two-sex population model of the dynamics of adults only. We assumed that the sex-specific (S , which could be either F(emale) or M(ale)) population size in the first year followed a Poisson distribution such that $N_{S,I,1} \sim \text{Poisson}(y_{S,I,1})$, in which $y_{S,I,0}$ is the sex-specific population count of number of adults on 15 November in the first year of study period (i.e., 1993), 87 for the females and 181 for the males. From the second year, we assumed that female and male population size was governed by population size in the previous year, survival, reproduction, and immigration such that

$$N_{F,I,t+1} \sim \text{Poisson} \left(N_{F,I,t} \times \left\{ \begin{array}{l} \varphi_{F,I,t} \times \varphi_{F,II,t} \times \varphi_{F,III,t} + \\ \gamma_t \times \tau_t \times \pi_t \times \varphi_{L,II,t} \times \varphi_{L,III,t} + \\ [y_t \times \tau_t \times (1 - \pi_t) + \delta_{F,I,t}] \times \varphi_{O,II,t} \times \varphi_{O,III,t} + \\ \delta_{F,III,t} \end{array} \right\} \right), \quad (1)$$

$$\text{and } N_{M,I,t+1} \sim \text{Poisson} \left(N_{M,I,t} \times \left\{ \begin{array}{l} \varphi_{M,I,t} \times \varphi_{M,II,t} \times \varphi_{M,III,t} + \\ \psi_t \times \gamma_t \times (1 - \tau_t) \times \varphi_{J,II,t} \times \varphi_{J,III,t} + \\ \delta_{M,III,t} \end{array} \right\} \right), \quad (2)$$

(see Supplementary Table S2 for a description and estimates of vital rates φ , γ , τ , π , δ and ψ). From this population model, we can estimate annual PGR $P_{S,t}$ as the change in population size N from one year to next for males or females such that

$$P_{S,t} = \frac{N_{S,I,t+1}}{N_{S,I,t}}. \quad (3)$$

In this model, we linked the population count data y with the population size N using a Poisson distribution such that

$$y_{S,I,t} \sim \text{Poisson}(N_{S,I,t}). \quad (4)$$

The probability of males being the dominant in their territory ψ was informed by the counts of dominant males $y_{D,I,t}$ among all males $y_{M,I,t}$ with a binomial model such that

$$y_{D,I,t} \sim \text{Binomial}(y_{M,I,t}, \psi_{I,t}). \quad (5)$$

Survival submodel. We used the Cormack-Jolly-Seber model⁵³ to estimate apparent survival over each of the three phases from individual capture-resighting data. The data were collated into an m-array format, which followed multinomial distributions. The cell probabilities of the m-array were calculated from stage-, phase-, and year-specific apparent survival probabilities and stage-, and phase-specific detection probabilities. We assumed that detection probabilities are constant across years due to intense survey effort (detection probabilities are very high, except for the dispersing juvenile females; Supplementary Table S3).

Recruitment submodel. We estimated fecundity γ_t using recruitment data (i.e., counts of adult females and different juveniles) such that

$$y_{JMF,II,t} \sim \text{Binomial} \left(y_{JMF,II,t} + y_{F,I,t}, \frac{\gamma_t}{\gamma_t + 1} \right), \quad (6)$$

in which $y_{JMF,II,t}$ is the sum of the counts of juvenile males and females. In our model, we assume that subordinate males do not contribute to reproduction, although in reality they can obtain substantial

paternity^{54,55}. However, the usurped paternity from dominant males does not affect the number of juveniles produced per group/female (i.e., fecundity), and hence we can ignore it in our analyses.

Additionally, we estimated the offspring sex ratio τ_t using a binomial model such that

$$y_{JF,II,t} \sim \text{Binomial}(y_{JMF,II,t}, \tau_t), \quad (7)$$

in which $y_{JF,II,t}$ is the counts of juvenile females (natal and non-natal juveniles combined). The female natal philopatry rate π_t was also estimated using a binomial model:

$$y_{L,II,t} \sim \text{Binomial}(y_{JF,II,t}, \pi_t), \quad (8)$$

in which $y_{L,II,t}$ is the counts of natal juvenile females.

Immigration submodel. We linked the immigration rates of females and males to immigration data. Female immigration rates were estimated in the recruitment phase $\delta_{F,I,t}$ such that

$$y_{GF,II,t} \sim \text{Poisson}(N_{F,I,t} \times \delta_{F,I,t}), \quad (9)$$

and in the scramble phase $\delta_{F,III,t}$ such that

$$y_{GF,I,t+1} \sim \text{Poisson}(N_{F,I,t} \times \delta_{F,III,t}), \quad (10)$$

in which $y_{GF,II,t}$ and $y_{GF,I,t+1}$ are counts of female immigrants at the end of recruitment phase and scramble phase (1 April and 15 November), respectively. Immigration rate of males were only estimated in the scramble phase $\delta_{M,III,t}$ such that

$$y_{GM,I,t+1} \sim \text{Poisson}(N_{M,I,t} \times \delta_{M,III,t}), \quad (11)$$

in which $y_{GM,I,t+1}$ is the counts of male immigrants.

Model implementation. We analysed the IPM in a Bayesian framework using Markov Chain Monte Carlo simulations, which were implemented in JAGS⁵⁶ using the R package jagsUI^{57,58}. We used vague priors Normal (0, 0.01) for all mean parameters, Gamma (0.01, 0.01) for all precision (inverse of variance) parameters, and Uniform (0, 1) for all detection probability parameters. By running three independent chains with different starting values for 10,000 iterations, with a burn-in of 8,000 iterations, we obtained 6,000 posterior samples without thinning. The convergence of model chains was checked using R-hat and Gelman-Rubin diagnostic statistics⁵⁹. The R-hat statistics for each parameter were well below 1.03, and thus the chains were well mixed.

Demographic pathway of climate impacts on population dynamics

To properly estimate the effects of climate on annual PGR, and the demographic pathway (via which vital rate) that influence annual PGR the most, we applied a retrospective path analysis²⁰ modified from a recent study²¹. Our retrospective path analysis model could account for uncertainty in the model inputs by fitting every linear model in the path diagram to each sample in the posterior distribution of each vital rate estimated by the IPM²¹. The model included paths between (a) climate/density variables to each relevant vital rate and (b) these vital rates to annual PGR (Supplementary Fig. 4). We estimated the coefficients of the first kind of paths in the same way as used by conventional path analysis. However, as annual PGR is completely determined by all vital rates, the path coefficients from vital rates to annual PGR were not estimated by the conventional path analysis, but derived mathematically from our population model (see below).

Density dependence. We considered the density-dependence of vital rates in both females and males, but in different ways. For the females, density-dependence is thought to act in the scramble phase and was incorporated by making the transition (survival) rate of females into a breeding position dependent on the number of vacant territories per female competing for such vacancy. Females that do not obtain a breeding position usually die and there are very few female super-numeraries in this population^{17,18}. The number of vacancies is associated with the number of adult females that died in the previous recruitment and non-breeding phase, while the number of females competing for vacancies includes natal and non-natal females that survived the non-breeding phase. Thus, we calculated the breeding vacancy availability index $n_{v,t}$ as $\frac{y_{F,I,t} - y_{F,III,t}}{y_{L,III,t} + y_{O,III,t}}$, in which y_t are annual population counts, and confined $n_{v,t} \leq 1$.

Additionally, early fledged juvenile females are more likely to settle in big breeding groups when they disperse¹⁷. More helper males in the territory in the scramble phase enhances the opportunity of obtaining a breeding position for those non-natal juvenile females by dividing the breeding group into two and splitting the territory. Thus, to consider the process of (breeding group) fission, we modelled the effect of male group size ($n_{g,t}$; i.e., the average number of males per female at the start of the scramble phase (1 August), which was calculated as $\frac{y_{M,III,t} + y_{J,III,t}}{y_{F,I,t} + y_{L,III,t} + y_{O,III,t}}$) on the survival rate of non-natal juvenile females in the scramble phase.

For the males, density-dependence is determined by the female dynamics, as the probability of males being the dominant in their territory ψ_t is driven by the number of territories/groups, which we assume is limited by the number of females. Therefore, we equated the number of dominant males to the number of dominant females, which implies that $\psi_t = \frac{y_{F,I,t}}{y_{M,I,t}}$, with $\psi_t \leq 1$. Finally, we assume that no other vital rates are affected by group size or density-dependent.

Effects of climate/density variables on IPM estimated vital rates. We tested the effects of relevant climate/density variables on each vital rate (Supplementary Fig. 1) by extending our previous IPM through including climate/density variables. To reduce the number of variables considered in the final retrospective path analysis model, we first conducted a variable selection procedure. We fitted a series of models, in which each vital rate (fecundity, sex- and phase-specific survival rates, and female phase-specific immigrations rates, see Supplementary Fig. 1) was fitted as a logistic or log-linear function of a single climate variable. The retrospective path analysis model only included a variable for a given vital rate if the 90% credible interval of the slope coefficient excluded 0²¹. The rationale for using a 90% instead of 95% credible interval for variable selection is that this reduces the chances of overlooking climate effects (false negatives). Although some climate/density variables were significantly inter-correlated (e.g., spring rainfall and breeding vacancy availability index; Supplementary Table S4), none of the equations for a given vital rate included climate/density variables that were correlated significantly at the alpha=0.05 level (Supplementary Fig. 4; Supplementary Table S4).

We modelled each estimated vital rate in the IPM with a mean α and a process error ϵ_t , where ϵ_t followed a Gaussian distribution with mean 0 and standard deviation σ . The process error estimates the among-year variance of a given vital rate that is not due to any modelled effects of climate/density variables (i.e., the fixed effects).

Stage- and phase-specific apparent survival rate ϕ_t was modelled using logistic regression with a logit link function such that

$$\text{logit}(\phi_t) \sim \alpha + \beta \times X_t + \epsilon_t \quad (12)$$

in which β denotes the estimated slope and X_t represents annual values of a given climate/density variable, or a combination of climate/density variables with β or X_t in different vectors.

Female phase-specific immigration rate $\delta_{F,t}$ or fecundity γ_t was modelled using Poisson regression with a log link function such that

$$\log(\delta_{F,t} \text{ or } \gamma_t) \sim \alpha + \beta X_t + \epsilon_t. \quad (13)$$

Path coefficients of vital rates to annual PGR. In our retrospective path analysis, the path coefficient of a given vital rate to annual PGR measures the contribution of the change of that vital rate to the change of annual PGR. This specific path coefficient can be approximated by the sensitivity of the annual PGR of the model to change in the vital rates. To quantify the sensitivity of annual PGR, we re-expressed annual PGR $P_{S,t}$ as (combining Eqs. (1)–(3)):

$$P_{F,t} = \left\{ \begin{array}{l} \phi_{F,I,t} \times \phi_{F,II,t} \times \phi_{F,III,t} + \\ \gamma_t \times \tau_t \times \pi_t \times \phi_{L,II,t} \times \phi_{L,III,t} + \\ [\gamma_t \times \tau_t \times (1 - \pi_t) + \delta_{F,I,t}] \times \phi_{O,II,t} \times \phi_{O,III,t} + \\ \delta_{F,III,t} \end{array} \right\} \quad (14)$$

$$\text{and } P_{M,t} = \left\{ \begin{array}{l} \phi_{M,I,t} \times \phi_{M,II,t} \times \phi_{M,III,t} + \\ \psi_t \times \gamma_t \times (1 - \tau_t) \times \phi_{J,II,t} \times \phi_{J,III,t} + \\ \delta_{M,III,t} \end{array} \right\} \quad (15)$$

and determined the partial derivative of $P_{S,t}$ with respect to each vital rate. For example, for male superb fairy-wrens, one unit change of immigration rate $\delta_{M,III,t}$ will contribute exactly one unit change of annual PGR $P_{M,t}$ (i.e., as $\frac{\partial P_{M,t}}{\partial \delta_{M,III,t}} = 1$ as can be seen from Eq. (15)). However, one unit change of adult male survival rate in the recruitment phase $\phi_{M,I,t}$ will contribute less, specifically $\frac{\partial P_{M,t}}{\partial \phi_{M,I,t}} = \phi_{M,II,t} \times \phi_{M,III,t} \approx 0.72$ (see Eq. (15) and Supplementary Table S2). This means that, if climate change decreases $\phi_{M,I,t}$ by 0.01 (1% lower survival), this will only lower annual PGR by 0.72%.

Future population extinction risk under climate change

We projected population dynamics under three climate change scenarios and contrasted these with the scenario of no further climate change to determine the impact of future anthropogenic climate change. For each climate change scenario, we specifically accounted for key sources of uncertainty: uncertainty among the 21 climate models, uncertainty in model parameter estimates, demographic and environmental stochasticity (both due to climate effects modelled explicitly and non-climate effects modelled as residual interannual noise). Demographic stochasticity reduces population growth rate at small population size⁶⁰, which is particularly relevant for studies quantifying risk of extinction.

As the first step, we extended our previous IPM by including the effects of influential climate/density variables on vital rates, as identified in the retrospective path analysis (Supplementary Fig. 4). Thereafter, we combined the climate and density-dependent IPM with forecasted climate variables to calculate the expected vital rates and population size year by year from 2022 to 2100. For each climate change scenario, we projected population dynamics in five steps: (i) we randomly selected one set of the forecasted climate variables from 21 global climate models, (ii) we sampled one set of vital rates, and the coefficients of each climate variable on the corresponding vital rate from 6000 sets of posteriors produced by the IPM; (iii) at each time we sampled from the previously estimated among-year variance of vital rates to incorporate the environmental stochasticity among years that is not due to the climate variable considered, and (iv) we ran the population model including demographic stochasticity and (v) repeated this procedure 100,000 times. For the scenario of no further climate change, the first step was different: we generated forecasted climate variables using Gaussian distributions with the mean and standard deviation of corresponding historical climate variables each time (i.e., 100,000 times).

Our model included density-dependence of vital rates by calculating breeding vacancy availability index year by year based on the projected population sizes in the relevant phases. For the offspring sex ratio τ , we assumed it was constant over time. Demographic stochasticity was incorporated by modelling population size as a random Poisson variable, which also ensures it is an integer and that extinction is clearly defined ($N = 0$). Our projection assumes that climate impacts on vital rates in the projection period will be the same as in our study period (e.g., no nonadditive effects of climate on vital rates), and there is no evolutionary rescue resulting in more adaptive vital rate responses to climate change.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data necessary to evaluate the conclusions of this study have been deposited in Zenodo with the link (<https://doi.org/10.5281/zenodo.18298289>). Daily weather for Canberra can be accessed from the Australian Bureau of Meteorology (<http://www.bom.gov.au/climate/data>). Forecasted climate data from global climate change models of the Coupled Model Intercomparison Project Phase 6 (CMIP6) are available at <https://ds.nccs.nasa.gov/thredds/catalog/catalog.html>.

Code availability

The code is available in Zenodo with the link (<https://doi.org/10.5281/zenodo.18298289>).

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Author contributions

L.L., L.E.B.K., A.C. and M.P. conceived and designed the study. A.C. and H.L.O. conducted the fieldwork and data management. L.L. performed the analyses with support from Q.Z., X.L., L.E.B.K., A.C. and M.P. L.L., A.C. and M.P. wrote the manuscript. L.L., Q.Z., Y.L., X.L., L.E.B.K., A.C. and M.P. contributed to the revisions of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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